This project will have two parts, creating an ER diagram and uploading the data.

ER Diagram

Use draw.io or other drawing software to create an ER diagram which includes every attribute in downloaded data in at least one table, you should not create new attributes, with 1 exception, you may create locationID to be a unique ID for location.
Each table should represent a real-world entity. Eventually the tables will need to be normalized, but that will wait for the next part of the project. You should have at least 3 tables, probably closer to 10-15 if done correctly.
Here is a guide to using this software to create ER diagrams.
https://drawio-app.com/blog/entity-relationship-diagrams-with-draw-io/ Either shapes with bubbles or tables are fine. YOU MUST USE THE DRAW IO SOFTWARE

You must use the crows foot/oracle notation we discussed in class, from the oracle link on the slides.

TO SUBMIT:
Submit a PDF file that contains the ER diagram you created.

Rubric:
0 points: diagram makes mistakes in cardinality or is missing some notation., or adds more attributes than locationID, or removes or adds more than 1-2 attributes
0.75 point: The diagram is mostly correct but has some major real issues such as missing/incorrect 1 cardinality notations, or 1-2 attributes.
1.25 points: Diagram is fully correct but uses wrong notation (not the one we learned in class) or other formatting errors

2 points: diagram is fully correct.

Downloading

Download the existing mortgage data from https://www.consumerfinance.gov/data-research/hmda/historic-data/
For this project we will want the following options:
All records Includes applications, denials, originations, institution purchases
Plain language labels and HMDA codes
For this project we will only use the New Jersey data.
Then download all the data for 2017 from NJ.

Please also click the links to the PDFs that help explain the data.

Uploading to data postgreSQL

You will need to upload the data to the ilab database. To do this, first log into ilabs then type the word "postgres"
to run a script that will open the postgres shell. You are already in a database coresponding to your netid.

You should produce a table with the correct type for each attributes, text for english sentences, int for numbers and sequence numbers etc. I would start by uploading it all as text, then casting into the correct type as needed. You will lose points if any of the types are not correct (for example)

You then need to write an SQL script to upload the CSV file into the ilab database. This will require creating a table first (call this table Preliminary), then running the script to upload the data. You can do this by putting the script in the same folder you run the postgres command from and then using \i to run the script.

You should change sequence_number into a primary key column with sequential numbers which will serve as the primary key for the new relation. You should not change the name of the column.

Lastly, you should write an SQL script that writes the contents of the database back to a CSV which is identical to the original CSV. You can use the diff command in linux to check to make sure that they are exactly the same.

TO SUBMIT:
Upload a series of screenshots of the results of "SELECT X,Y,Z FROM Preliminary;" where X,Y,Z are names of attributes. You should upload as many as are needed to show all attributes.

An SQL script that contains the SQL commands to generate the initial table and fill it from the CSV, including the sequence number logic.

An SQL script to convert back to the original CSV, including removal of the sequence number.

We also may look at the actual ilab database to check that the table is there

RUBRIC:
0 points: Code does not run on the ilab machine, or is missing some of the steps in the assignment.
1 points: Code mostly works but has some major substance issues, for example some of the types are incorrect, or we lose some precision on the floating point numbers.
1.5 point: Code runs and works mostly but has cosmetic issues, for example maybe the rows of the reconstructed CSV are in the wrong order, or Sequence number stays in the new CSV.
2 points: code is fully correct.

Academic Dishonesty

Code copied from anywhere will be dealt with according to the Rutgers Academic Dishonesty policy.
If the copied code's source is referenced, the maximum penalty will be 75% off the given assignment,
and likely no penalty at all depending on how much was copied and whether you had the license.

It is OK to take code from official language documentation, but you should still cite it.

AI is permitted for this assignment. However, as a result your code must fully comply with ALL parts of the assignment or significant penalty will be taken.

For this project in particular since there is so little code, code copied with citation from any source will almost always be fine.

If the code is not cited, and makes up a substantial portion of the project, the MINIMUM penalty will be a zero on the assignment.

README file

You must also submit a README file (named README.txt) with clearly
delineated sections for the following.

0. Please write down the full names and netids of both your team members.
1. Are there known issues or functions that aren't working currently in your
   attached code? If so, explain. (note that you will get half credit for any reasonably sized bug that is fully explained in the readme)
2. Collaboration: Who did you collaborate with on this project? What resources and references did you consult? Please also specify on what aspect of the project you collaborated or consulted.

3. What useful insights can you get from this data? What rules did you use for dividing up the attributes into entities?
4. What problems did you face developing code for this project? Around how long did you spend on this project (This helps me decide what I need to explain more clearly for the next projects)
5. Which team member has stored the data in their database for grading purposes/